

Shahziar Karim Safit

EEE Graduate — Wireless Communication & 6G Security Researcher

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CAREER OBJECTIVE

Electrical and Electronic Engineering graduate from Rajshahi University of Engineering Technology (RUET) with broad research interests in Wireless Communications, 6G Networks Security, Antenna RF Engineering, Machine Learning, VLSI Semiconductor Technology, and Solar Cell Photovoltaic Technology. Experienced in research and technical projects involving RIS- and relay-assisted wireless communication systems, physical-layer security, diversity techniques, and wireless performance analysis. Seeking opportunities to pursue research, develop technical expertise, and contribute to innovative solutions across diverse areas of Electrical and Electronic Engineering.

EDUCATION

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|--|-------------------------------------|
| • Rajshahi University of Engineering & Technology (RUET)
<i>B.Sc. in Computer Science & Engineering, CGPA: 3.66 / 4.00</i> | Rajshahi, Bangladesh
2022 – 2026 |
| • Notre Dame College
<i>Higher Secondary Certificate (HSC), GPA: 5.00 / 5.00</i> | Dhaka, Bangladesh
2020 |
| • Ideal School and College
<i>Secondary School Certificate (SSC), GPA: 5.00 / 5.00</i> | Dhaka, Bangladesh
2018 |

RESEARCH INTEREST

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|---------------------------|--------------------|--------------|
| • Wireless Communications | • Antenna Design | • VLSI |
| • 6G Networks & Security | • Machine Learning | • Solar Cell |

RESEARCH PAPERS

- **Secure Outage Performance Analysis of Reconfigurable Intelligent Surface-Aided Networks with Selection Combining Diversity Scheme** *1st Author* - International Conference on Power, Electronics, Communications, Computing, and Intelligent Infrastructure 2026 (PECCII), Pabna, Bangladesh - *Accepted*
- **Secure Outage Performance Analysis of Reconfigurable Intelligent Surface Assisted Communication System Over κ - μ Shadowed Fading Channels** *2nd Author* - 5th IEEE International Conference on Signal Processing, Information, Communication and Systems (SPICSCON 2026), BAUET, Qadirabad, Natore, Bangladesh - *Accepted*
- **Ergodic Secrecy Capacity Analysis of Reconfigurable Intelligent Surface-Assisted Wireless Networks with SC and MRC Diversity Schemes** *1st Author* - 29th International Conference on Computer and Information Technology (ICCIT 2026), Cox's Bazar, Bangladesh - *Submitted*
- **Comparison Between the Secrecy Capacity of Reconfigurable Intelligent Surface Aided and Relay-Assisted Wireless Communication Systems in the Presence of Multiple Eavesdroppers** *2nd Author* - 14th International Conference on Electrical and Computer Engineering (ICECE 2026), BUET, Dhaka, Bangladesh - *Submitted*
- **Adaptive Transfer Learning with EfficientNetB1 for Multi-Class Brain Tumor Identification in MRI Images** *3rd Author* - 29th International Conference on Computer and Information Technology (ICCIT 2026), Cox's Bazar, Bangladesh - *Submitted*

RESEARCH EXPERIENCE

- **Undergraduate Thesis** Jan. 2025 – Aug. 2026
Research Project — Undergraduate Thesis
 - **Contribution:** Developed a comprehensive analytical framework for RIS-assisted wireless networks with SC and MRC diversity, deriving closed-form expressions for ESC, PNZSC, IP, and SOP.
 - **Analysis:** Investigated the effects of RIS elements, antenna configurations, SNR, and diversity schemes on physical-layer security.
- **Secure Outage Performance Analysis of RIS-Assisted Networks**
Research Project — First Author
 - **Contribution:** Derived closed-form expressions for PNZSC, IP, and SOP in RIS-assisted wireless networks employing SC diversity over Rayleigh fading channels.
- **RIS-Assisted Communication over κ - μ Shadowed Fading**
Research Project — Second Author
 - **Contribution:** Derived series-form expressions for PNZSC, IP, and SOP for RIS-assisted communication over κ - μ shadowed fading and investigated the effects of κ , μ , and RIS elements.
- **Ergodic Secrecy Capacity of RIS-Assisted Wireless Networks**
Research Project — First Author
 - **Contribution:** Derived closed-form ESC expressions for RIS-assisted networks with SC and MRC diversity and conducted a comparative analysis of their secrecy performance.
- **Secrecy Capacity of RIS-Assisted and Relay-Assisted Systems**
Research Project — Second Author
 - **Contribution:** Developed a comparative analytical framework for the ESC of RIS-assisted and relay-assisted systems in the presence of multiple eavesdroppers.
- **Multi-Class Brain Tumor Identification from MRI Images**
Research Project — Third Author
 - **Contribution:** Contributed to a reproducible EfficientNetB1 transfer-learning pipeline for four-class brain tumor MRI classification, achieving 98.00% test accuracy and 98.00% macro F1-score.

PROFESSIONAL EXPERIENCE

- **Industrial Training** Mar. 2, 2025 – Mar. 9, 2025
Siliconova Limited (Semiconductor & VLSI Technology Company)
 - Gained practical exposure to ASIC design flow, RTL coding using Verilog/SystemVerilog, CMOS and VLSI design concepts, FPGA and semiconductor technologies, and digital circuit design and verification.
 - Worked with Verilog RTL coding and simulation, ASIC physical design, Design for Testability (DFT), timing analysis, synthesis flow, and Cadence IC design tools.

TEACHING EXPERIENCE

- **QNA Teacher and Script Evaluator** Apr. 2023 – Aug. 2026
UDVASH — Part-time
 - **Teaching:** Provided academic guidance and conceptual support to students through question-answer-based teaching.
 - **Script Evaluation:** Evaluated students' scripts and provided constructive feedback to improve their academic performance.
- **Private Tutor**
SSC & HSC Students
 - **Teaching:** Provided one-to-one academic instruction and conceptual guidance to SSC and HSC-level students.
 - **Mentoring:** Assisted students with problem solving, examination preparation, and strengthening fundamental concepts.

PROJECTS

- ESP32-Based Interactive Quadraped Robot with Wireless Motion Control** Mar. 2026 – Jun. 2026
Capstone Project 2
 - Developed an 8-DOF quadraped robot using ESP32, eight servo motors, Wi-Fi-based web control, OLED facial animation, and a 3D-printed mechanical structure.
- IoT-Based Automatic Accident Detection and Rescue System** Jun. 2025 – Aug. 2025
Capstone Project 1
 - Designed an IoT-based accident detection system using ESP32, accelerometer, and GPS for real-time impact detection, location tracking, and emergency alerts.
- IoT-Based Intelligent Soil Moisture-Controlled Watering System** Sep. 2024 – Oct. 2024
Electrical Shop Practice
 - Developed an IoT-based automated irrigation system using Arduino, ESP8266, soil-moisture sensors, and a water pump for real-time soil monitoring and automatic plant watering.
- Line Follower and Obstacle Avoidance Robot** Jun. 2024 – Jul. 2024
Academic Project
 - Built a mobile robot capable of autonomous line following and obstacle avoidance using sensors and microcontroller-based motor control.

TECHNICAL SKILLS

- | | |
|---|--|
| 1. Simulation Software: | MATLAB, Simulink, Mathematica, CST Studio Suite, Multisim |
| 2. CAD Software: | AutoCAD, SolidWorks, EAGLE PCB, Microsoft Visio |
| 3. Programming Languages: | C, C++, Python, Assembly Language |
| 4. Hardware Skills: | ESP32, Arduino, ESP8266, FPGA, Microcontrollers |
| 5. Electronics & Circuit Design: | Proteus, MicroWind, LOGO! Soft Comfort |
| 6. Others: | LaTeX, Overleaf, VS Code, Microsoft Office (Word, Excel, PowerPoint) |

ADDITIONAL INFORMATION

- Languages:** Bengali (Native), English (Professional Working Proficiency)

REFERENCES

- Dr. Ajay Krishno Sarkar:** Professor, Department of Electrical & Electronic Engineering, Rajshahi University of Engineering & Technology (RUET). Phone: +880-1792202808. Email: aksarkar@eee.ruet.ac.bd; sarkarajay139@gmail.com.
- Dr. Md. Zahurul Islam Sarkar:** Professor, Department of Electrical & Electronic Engineering, Rajshahi University of Engineering & Technology (RUET). Phone: +880-1789538712, +880-2588867288. Email: islam89118@gmail.com.

DECLARATION

I hereby declare that all the information stated above is true and correct to the best of my knowledge and belief.

Safit

Shahziar Karim Safit